Oral Radiology 25-26

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GOOD MORNING

PANORAMIC RADIOGRAPHY

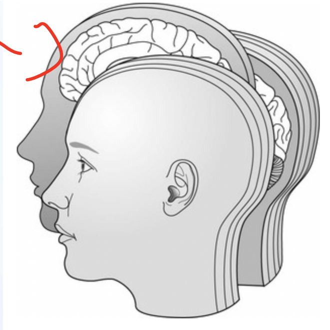
OPG

By the end of the lecture student should be able to :

1. Describe the working principle of tomographic and Panoramic radiography

2. Enumerate the causes for Panoramic radiographic faults

-Tomography is a radiographic technique that allows the imaging of one layer or section of the body while blurring images from structures in other planes.



-OPG – OrthoPantomoGram/ Pantomography

Panoramic meaning 'an unobstructed view of a region in every direction'.

OPG- It is a radiographic technique that produces a single tomographic image of the facial structures, including both maxillary and mandibular arches and their supporting structures.

Final MAQ

Indications for Panoramic Radiography

1

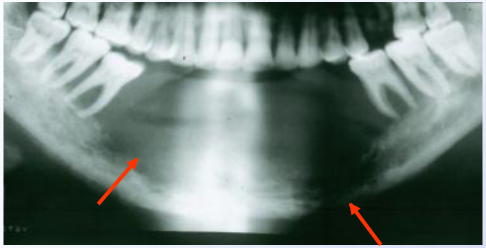
- When large anatomic coverage is required to study lesions .

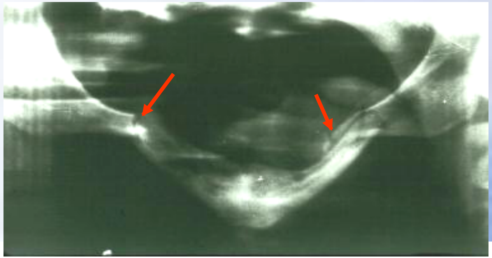
2

- Pathology:- Developmental anomalies, cysts ,tumors , impacted teeth

3

- Detection of fractures, follow up of treatment.





Indications for Panoramic Radiography

first

4)
- Mixed dentition analysis

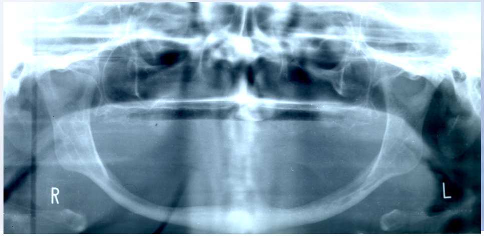
5)
- Alveolar bone levels in periodontal disease

6)
- Implant placement

7)
- Temporo-mandibular joints and their disorders.

-

Q) for anterior teeth what method is contraindicator?
A) OPG, because spine and vertebra bone



Adenturist

Advantages of Panoramic Radiography

final

1. Coverage of the entire maxillary and mandibular dentition on a single film.
2. Ease of technique
3. Patient acceptance is high.
4. Less time consuming.

Disadvantages of Panoramic Radiography

Final

1. Image quality- Inadequate sharpness and detail (Poor)
2. Magnification of images
3. Geometric distortion.
4. Overlapping of images.
5. Soft tissue and air shadows (can't see many clearly)
6. Ghost or artefactual shadows

- **Digital panoramic machines** come with sensor and appropriate software.

-The receptor/sensor used: what are the sensor used?

  -Charged Couple Device(CCD) or

  -Complementary Metal Oxide Semiconductor Sensors(CMOS).

- The sensor array transmits -electronic signal to- computer –

- displays the image on the view screen as it is being acquired.

- Allows to modify the image Characteristics- eg: Contrast and Density adjustments.
- Exporting the digital image in DICOM format.

- DICOM is a standard that specifies handling, storage, printing, and transmission of medical images

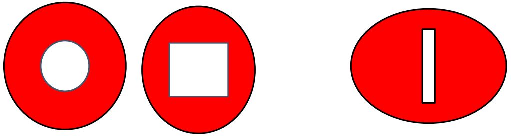
The main components of Panoramic unit include:

- X-ray tube head
- Head positioner
- Exposure controls

Final

Q) what defferences between Intraoral and Panoramic rad.?

- **Intraorally**- round or rectangular collimator.
- **Panoramic radiography**-
- TWO Collimators
- Narrow, rectangular .



intraoral
one collimation

panoramic
two collimation

2 collimators in opg

Exposure Settings

Final

- The kVp and mA can be adjusted on panoramic machines.
- **Exposure time cannot be changed.**

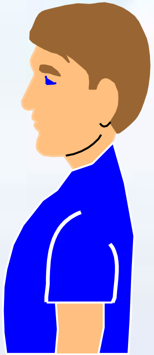
- Appropriate exposure setting should be selected, typically in the range 60-80 kVp .

- The exposure in panoramic radiographs is

  "9-26 μSv"

Patient Preparation

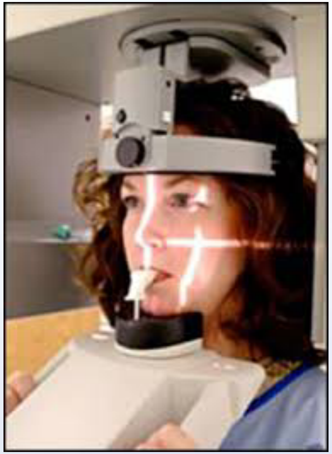
1- Ask the patient to remove glasses, jewelry, dentures, hearing aids,



2- Place the lead apron on the patient (no thyroid collar; it might block part of the x-ray beam).

The patient is encouraged to stand straight

- Head positioning – Light beam marker positioning guides (3 guides).



- The mid-sagittal plane perpendicular to the floor.
- The Frankfurt plane roughly parallel to the floor.

3 indicators:

1. Vertical-
2. Horizontal
3. Canine

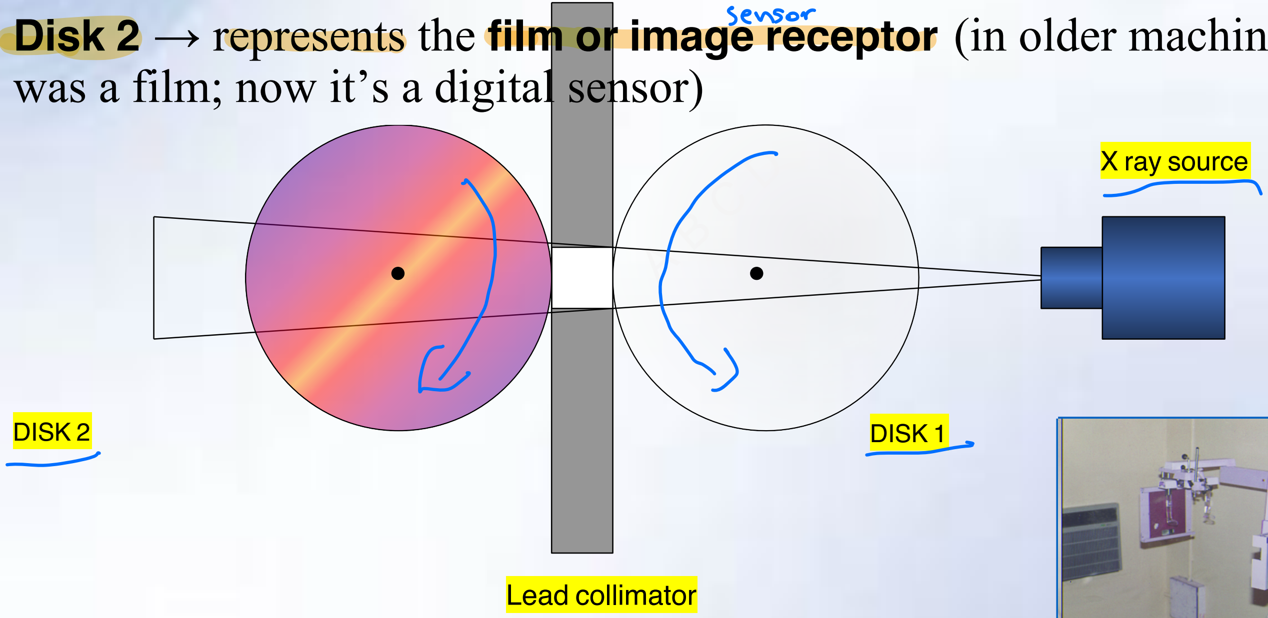
- The maxillary and mandibular central incisors placed edge to edge in the grooves of a bite block.

- As **Disk 1 (X-ray tube)** moves in one direction,
- **Disk 2 (receptor)** moves in the opposite direction.
- This synchronized movement keeps a **narrow focal trough (image layer)** in sharp focus.

Basic Principle of Panoramic Radiography

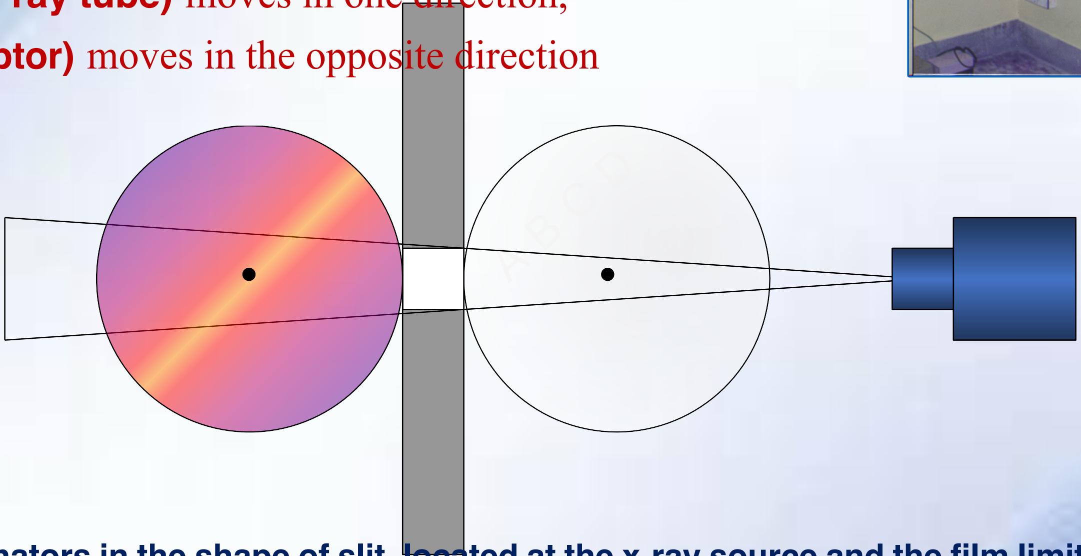
- **Two Rotating Components**
- **Disk 1** → represents the **X-ray tube head (source of X-rays)**
- **Disk 2** → represents the **film or image receptor** (in older machines it was a film; now it's a digital sensor)

sensor



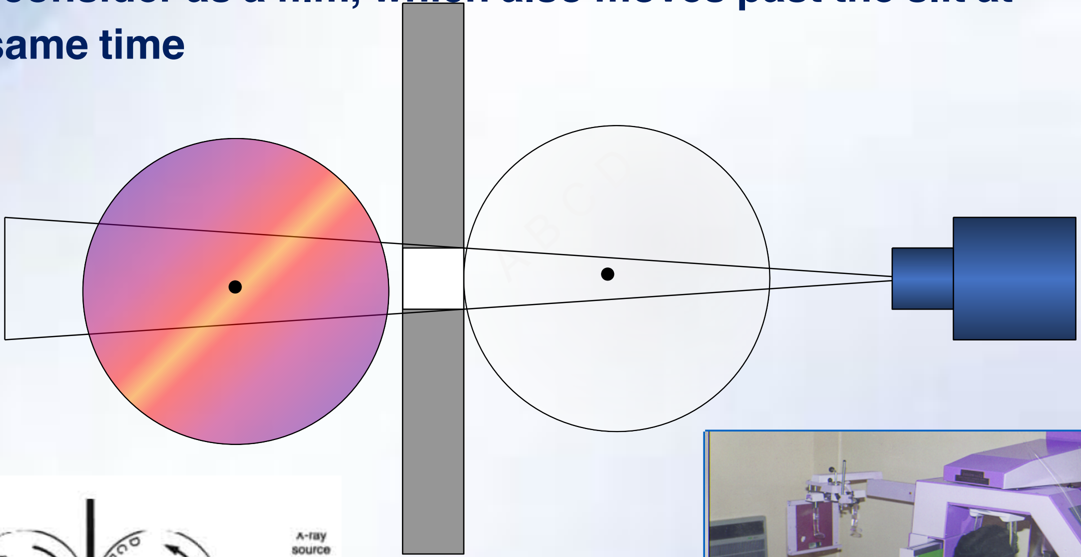
- **Consider two adjacent disks rotating at the same speed in opposite direction, as an x-ray beam is directed through the center of rotation of the two disks.**

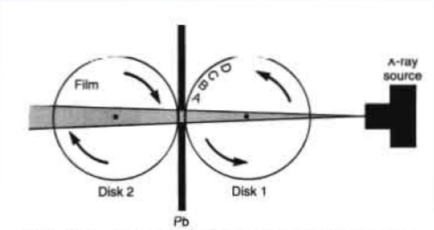
- As **Disk 1 (X-ray tube)** moves in one direction,
- **Disk 2 (receptor)** moves in the opposite direction



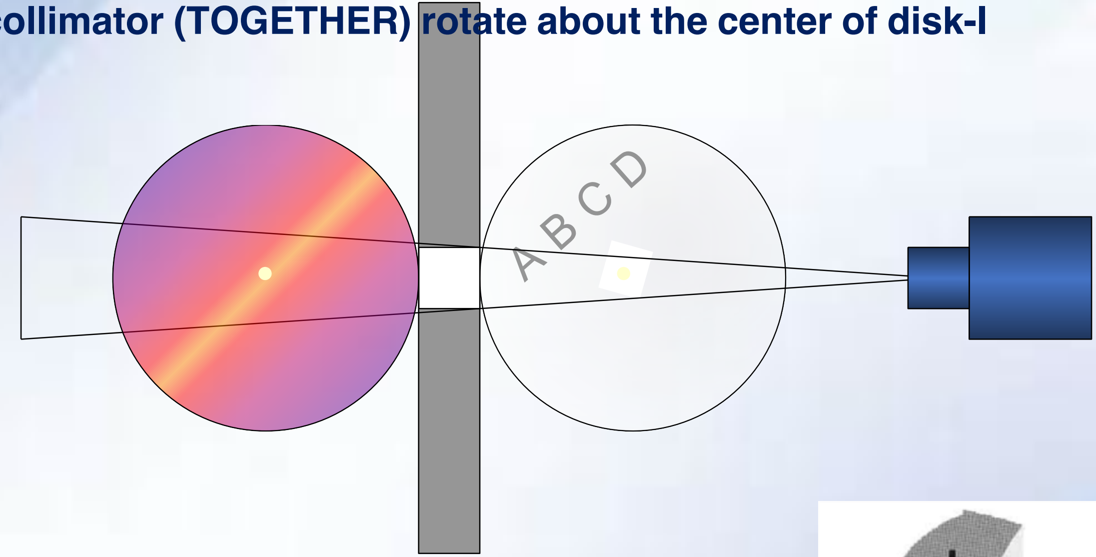
- **Lead collimators in the shape of slit, located at the x-ray source and the film limits the central ray to narrow slit beam**
- **This synchronized movement keeps a narrow focal trough (image layer) in sharp focus.**

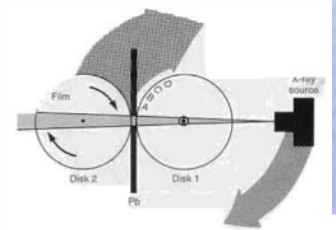
♪ **The radiopaque objects A,B,C,& D on disk-1 rotate past the slit and their images are recorded on disk-2 which we now consider as a film, which also moves past the slit at the same time**





- **The same relationship of moving film to image is achieved if the x-ray source is rotated so that the central ray passes through the center of rotation of disk-1 as well as disk-2, and the film collimator (TOGETHER) rotate about the center of disk-I**



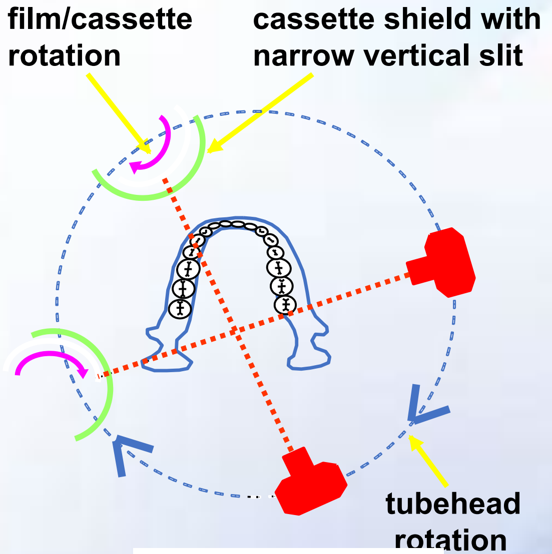


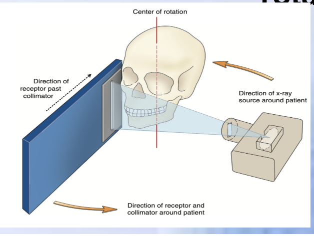
Tubehead Rotation

The X-ray tube rotates around the patient's head in one direction while the film rotates in the opposite direction. 223

The x-ray beam passes through a narrow vertical opening in the cassette shield.

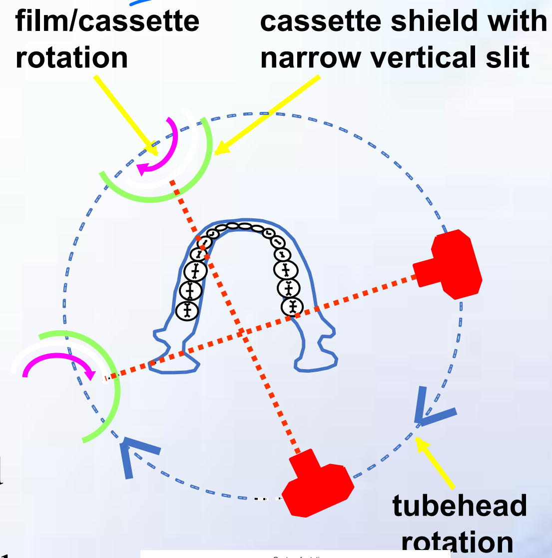
The cassette rotates within this shield, constantly exposing different parts of the film as the whole unit rotates.



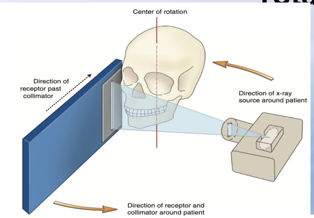


Tubehead Rotation

The cassette rotates within this shield, constantly exposing different parts of the film as the whole unit rotates.

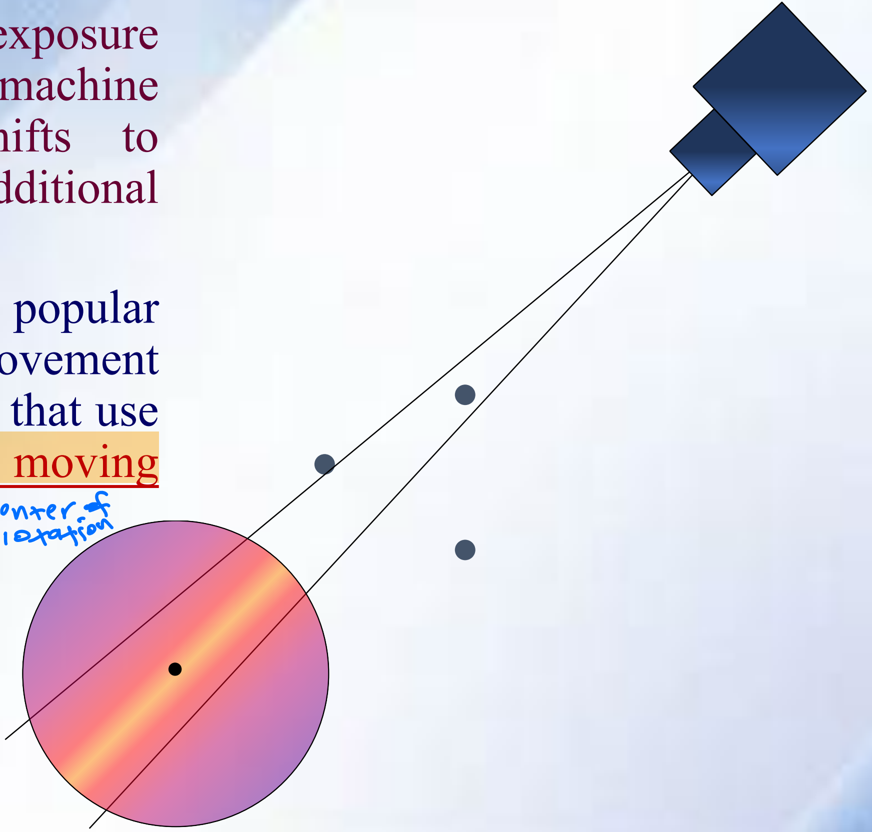


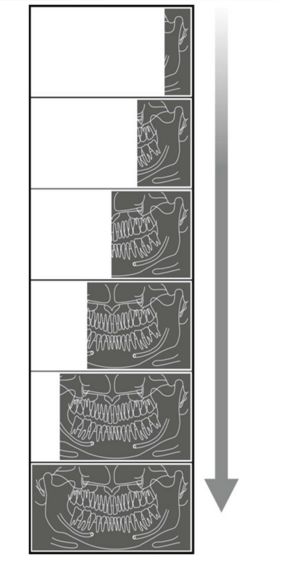
The **center of rotation** (between Disk 1 and Disk 2) **keeps changing** during the exposure.(due to curved shape of the dental arches)

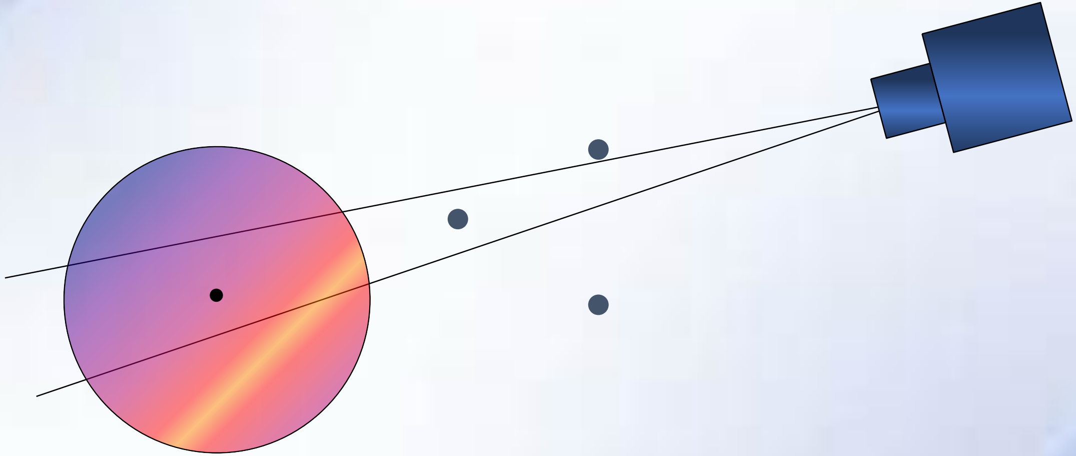
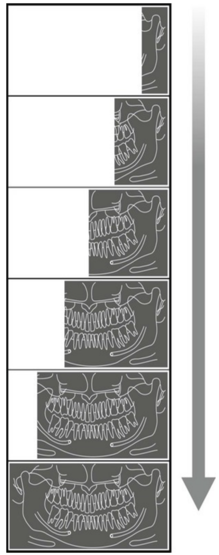


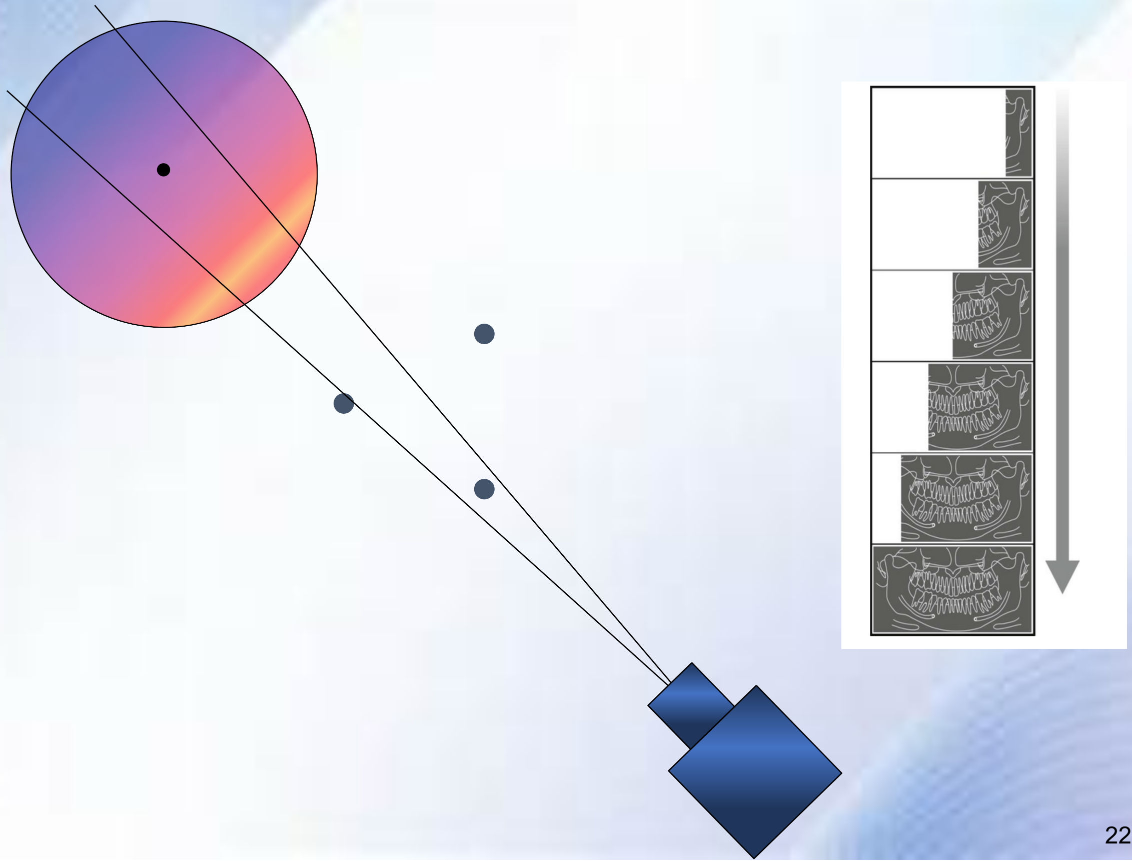
During the exposure cycle, the machine automatically shifts to one or more additional rotation centres.

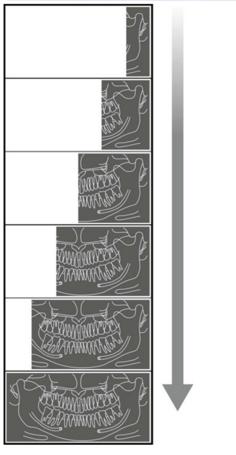
The most popular mechanical movement patterns are those that use a continuously moving rotation centre. center of rotation

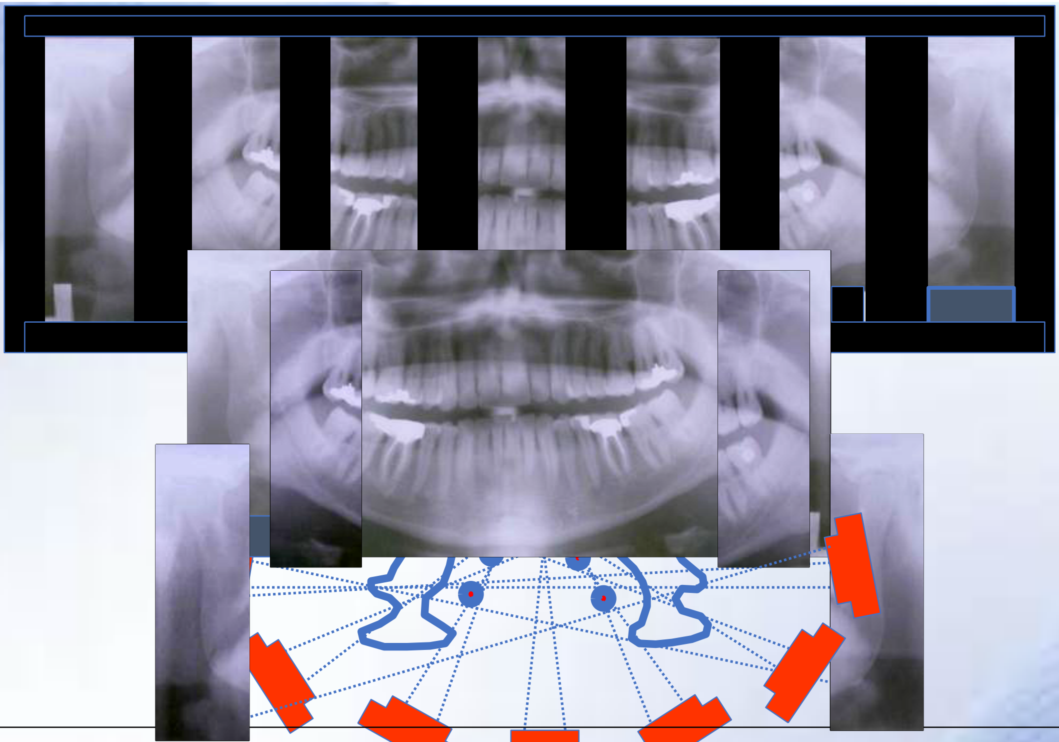




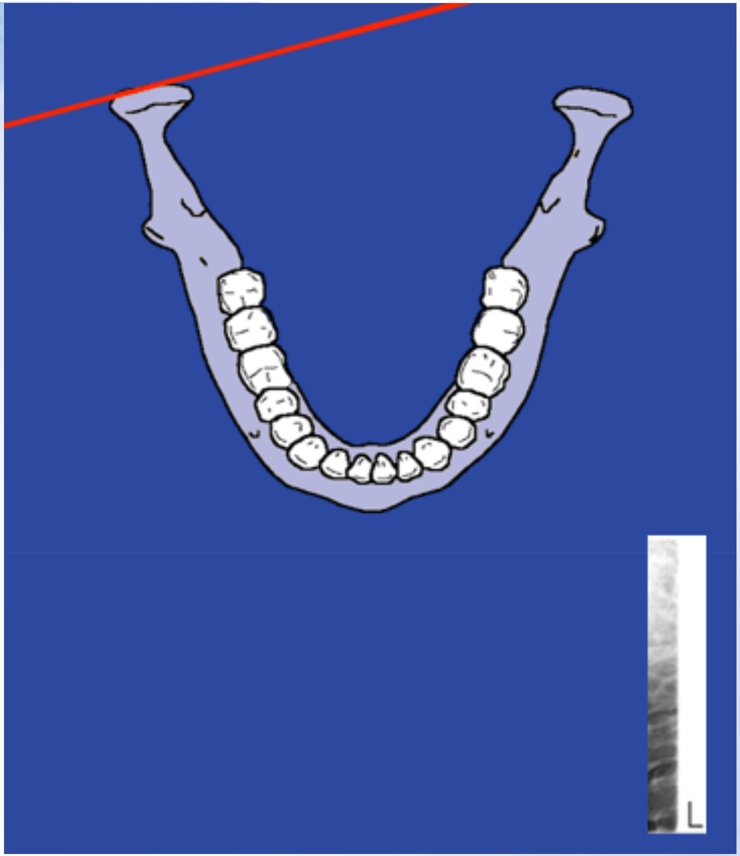









As the tubehead rotates around the patient, the x-ray beam passes through different parts of the jaws, producing multiple images that appear as one continuous image on the film ("panoramic view").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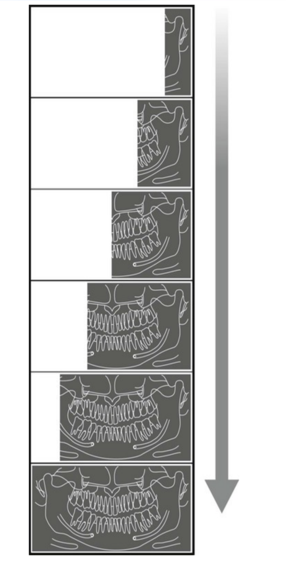


Focal Trough (image layer)

Q) what is focal through?

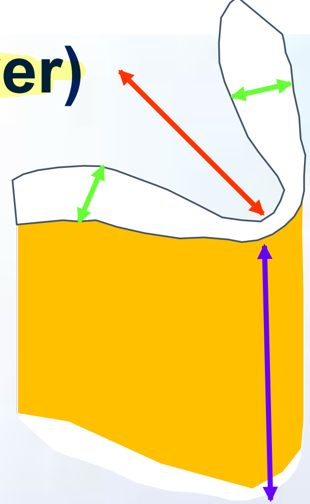
The Focal Trough is a Three-dimensional curved zone or image layer in which structures are reasonably well defined.

very clear

Machines are designed to have zone, shaped like dental arches- this zone corresponds to the shape of the upper and lower jaws. **(*HORSE SHOE SHAPED*)**

what is the shape of focal trough?

- The closer the rotation centre is to the teeth, the narrower the focal trough in that area .



Final #

The three dimensions of the focal trough are:

1. Front-to-back (anterior-posterior). Red arrow.

2. Side-to-side (Bucolingual). Green arrows.

3. Up-and-down (Vertical). Blue arrow.

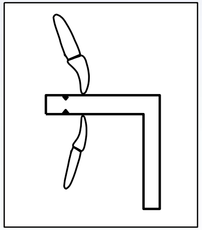
Teeth Too Anterior

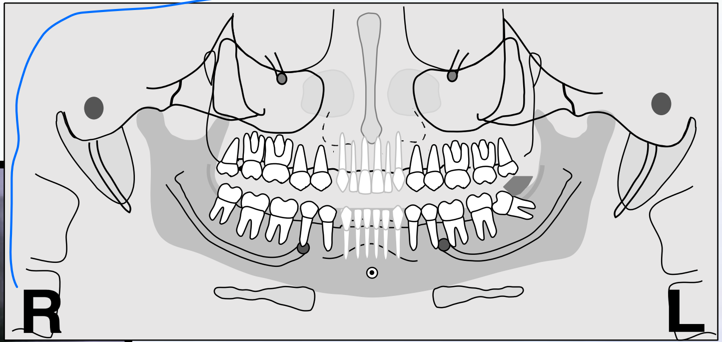
Final !

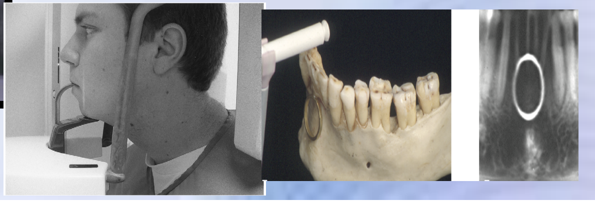
Q) what happend

When the object is positioned anterior to the Focal trough ?-

A) blurring and narrowing of the anterior teeth.





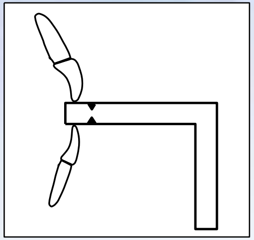


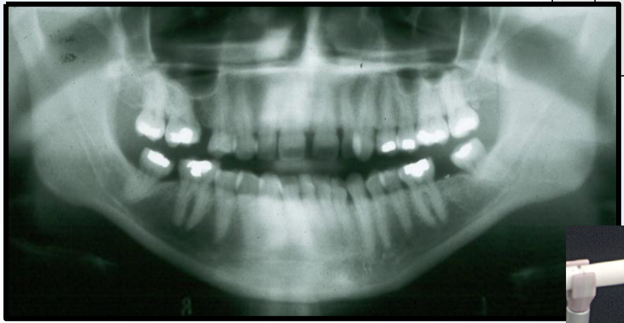
Teeth Too Posterior

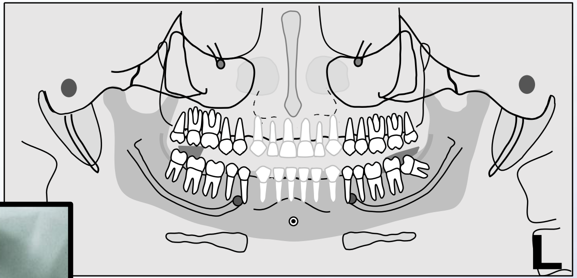
Final

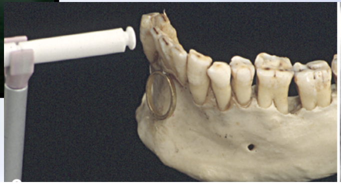
what happend

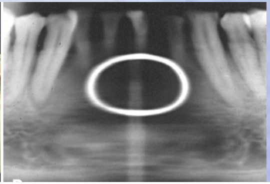
When the object is positioned posterior to Focal trough- ? A) image of structures is elongated horizontally (increase width) on the film.







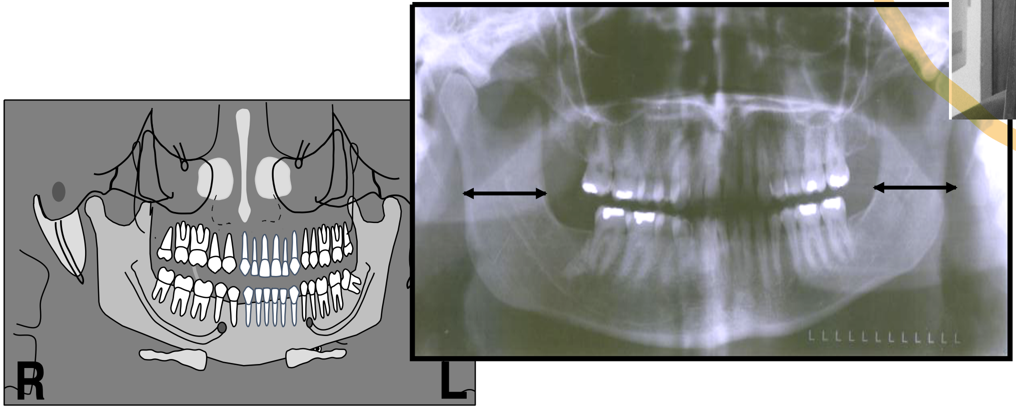




Final

Head Turned

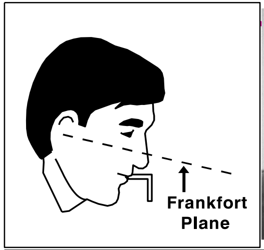
The teeth are smaller on the side to which the head is turned.

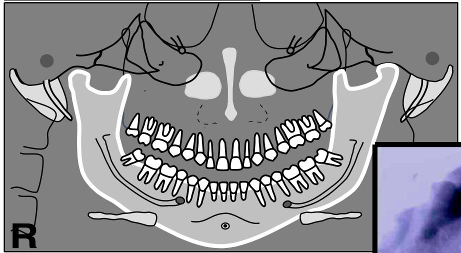


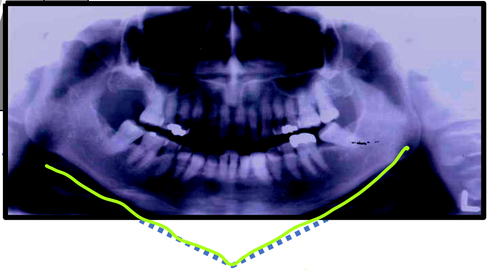
OSPE #

Final

Head Tipped Down



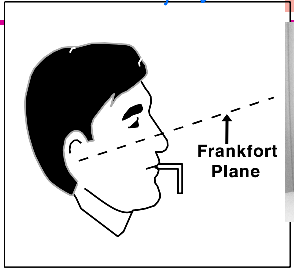


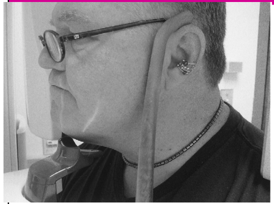


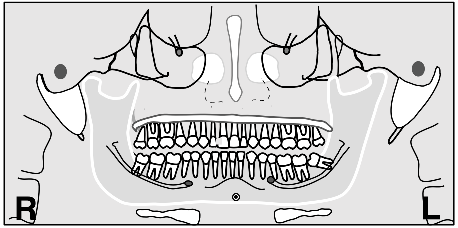
V-shape

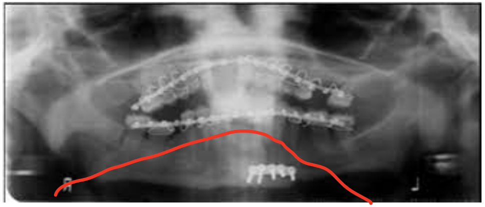
"**SMILING**" APPEARANCE

Head Tipped Up









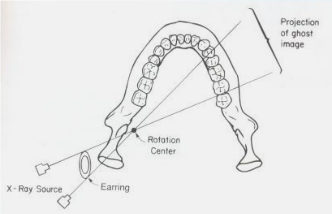
FLAT, REVERSED"

SMILING"DOWNWARD "

Final

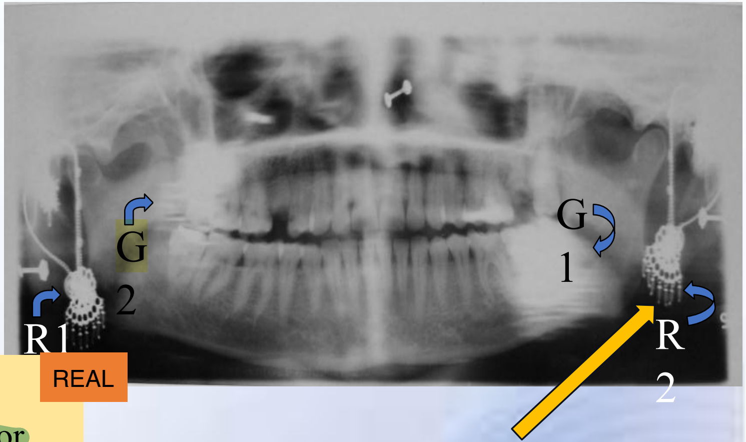
Formation of what Ghost image

Ghost image is formed when the object is located between x – ray source and center of rotation.



what **Ghost images characteristics:** ? SAQ

- Appears on opposite side of film (ear ring)
- Has same morphology as its real counterpart
- Appears higher up on radiograph
- More blurred than real image
- Ghost image are always REVERSED, with left to right being shifted



- Examples include:

Anatomical structures- Hyoid bone, cervical spine, inferior border of mandible, posterior border of ramus, condyles.

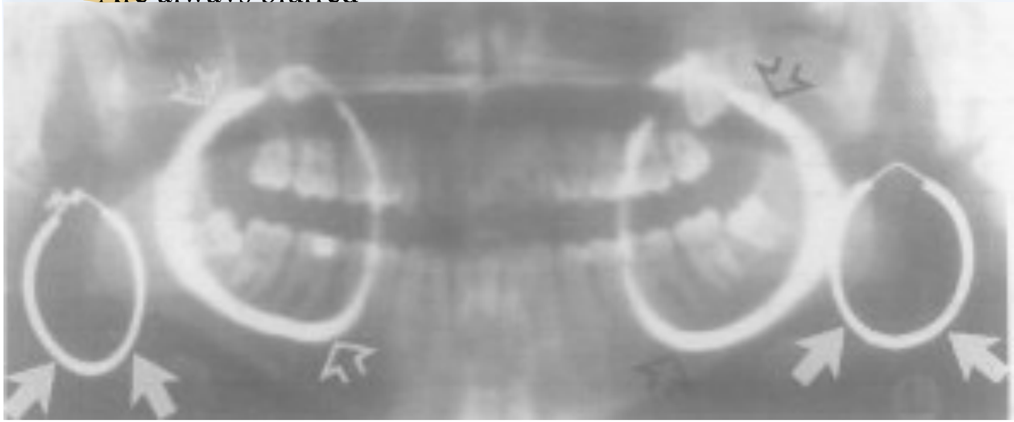
- **Objects**: Chin rest, right and left markers, earrings, neck chains.

- **Double image** is a pair of real images formed by an object lying within central portion of oral and maxillofacial region which are intercepted twice by the beam.

Final

CHARACTERISTICS:

- One image is the mirror image of other
- Both images are real images
- Each image will have same location on the opposite side
- Double images only occur with midline objects falling in diamond shaped zone in midline
- Are always blurred



- DOUBLE IMAGE ON SAME SIDE
- GHOST ON OPPOSITE SIDE

Reference:

White and Pharoah ,Oral Radiology- Principles & interpretation edt 7, chap 10 pg 166-184.